

devices operate at a frequency of up to 60MHz with a flash accelerator to obtain maximum efficiency. The accelerator provides up to 128kB of embedded flash memory for code/data storage and 8kB of embedded SRAM memory for system operation and application program usage.

Holtek Semiconductor India Pvt Ltd
www.holtek.com

Digital signal controllers

System developers designing high-end embedded control applications need flexible options to provide scalability as projects increase in complexity. To meet these needs, Microchip Technology has announced new dual- and single-core dsPIC33C digital signal controllers (DSCs) with options to meet changing application requirements across memory, temperature and functional safety. The dsPIC33CH512MP508 dual-core DSC enables support for applications with larger program memory requirements, whereas the dsPIC33CK64MP105 single-core DSC adds a cost-optimised version for applications that require smaller memory and footprint.

Microchip Technology Inc.
www.microchip.com

DC/DC converters

Traco Power has expanded its THN 15/20/30 WI series of 15W, 20W and 30W DC/DC converters in the standard $2.54 \times 2.54 \times 1.06$ cm ($1 \times 1 \times 0.4$ -inch) package with ultra-wide 4-1 input ranges. The models with -A1 suffix feature an extended trim range up to 6V DC. The extended trim function up to 6V DC makes these converters perfect for LDO regulator applications, reducing power losses over LDO while eliminating almost all ripple and noise components, and provides a clean 5V DC output voltage. This makes these converters ideal for providing isolation and regulation for circuits in low-noise applications.

Traco Power
www.tracopower.com

Thyristor

Littelfuse has introduced the first Pxxxx-S4xLRP series SIDACtor protection thyristor, a robust solution for protecting composite video blanking sync signal lines and ports from damaging over-voltage transients. The combination of 6V operating voltage and 100A 5/310µs surge peak current capability and a low junction capacitance rating make P0080S4BLRP ideal for protecting a wide range of equipment that need data line protection. The PxxxxS4xLRP series provides a compact, surface-mount solution, which simplifies designing it into PCB layout.

Littelfuse Inc.
www.littelfuse.com

Incremental encoder

With their exceptional low-speed performance, stepper motors are often a good choice where a servo is required but a gearbox for some reason is not beneficial. With their high pole count, stepper motor servos work better with high-resolution encoders. TMCS-28 is a low-cost, small-size optical incremental encoder for use with stepper motors and three-phase PMSM/BLDC motors. It comes with high-resolution optical code wheels with a resolution of up to 10k lines.

Trinamic Motion Control GmbH & Co KG
www.trinamic.com

Low-power DC/DC converters

R0.5ZX and R1ZX are based on 2kV DC isolated low-profile R1SX SMD series but include an internal linear regulator to give a precise, load-independent and low noise output. The already low output ripple and noise can be reduced to 2mVp-p with a simple external filter circuit. Output is also continuously short circuit protected. In the event of continuous overload or over-temperature condition, output shuts down, thus protecting the converter and the application from over-current damage. Output also automatically restarts once the fault condition has been lifted.

RECOM Power
www.recom-power.com

Microcontroller

Nuvoton Technology has announced NuMicro-enhanced 1T 8051-based MS51 series MCU. NuMicro MS51 series provides flexible internal clock system adjustable from 10kHz to 24MHz operating speed, 24mA high driving capability for each I/O pin, and high immunity such as 8kV ESD and 4.4kV EFT to ensure reliability even in harsh conditions. It operates at voltage range 2.5V ~ 5.5V and $-40^{\circ}\text{C} \sim +105^{\circ}\text{C}$ industrial temperature grade. The series offers a range of memory with 8/16/32kB flash memory and 1/2kB SRAM, and provides packages types from 10 to 32pins.

Nuvoton Technology Corp.
www.nuvoton.com

Isolated DC-DC converter

Key features of URH_LP_20WR3 converter are:

- Ultra-wide 4:1 input voltage range
- High efficiency up to 89 per cent
- No-load power consumption as low as 0.12W



- Operating ambient temperature range: $-40^{\circ}\text{C} \sim +85^{\circ}\text{C}$
- Reinforced insulation, I/O isolation test voltage 5kV AC, rated for 250VAC working voltage
- Transformer creepage 8mm, transformer clearance 8mm
- Low leakage current $< 5\mu\text{A}$
- Input under-voltage protection, output short-circuit, over-current, over-voltage protection

Mornsun Guangzhou Science & Technology Co. Ltd
www.mornsun-power.com

TEST & MEASUREMENT

Vector signal generator

Keysight Technologies has launched the first dual-channel microwave signal generators that support up to 44GHz signals and 2GHz radio frequency (RF) modulation bandwidth in

